ModelIt!

Student Activity Pack

Why Does Ice Cream Melt Faster on a Hot Day?

How Heat Energy Transfers from the Environment to Frozen Treats

Grade 05L2 | NGSS 5-PS1-1, 5-PS1-3 | Cause and Effect

STUDENT NAME _____ DATE _____

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why does ice cream melt? What causes it to change from solid to liquid?

2 If you hold an ice cube in your hand, your hand feels cold. Is cold moving to your hand, or is something else happening?

3 Draw what you think is happening to the ice cream particles as it melts.

4 What could you do to keep ice cream frozen longer without a freezer?

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Air Temperature | Container Insulation | Heat Transfer Rate | Melting Speed | Ice Cream State

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Hot Summer Day

High air temperature, low insulation

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Insulated Container

High air temperature, high insulation

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Cool Day, No Insulation

Low air temperature, no insulation

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Ice Cream Volume — Larger scoops take longer to melt because there is more frozen material to warm up

Wind Speed — Moving air carries heat to the ice cream faster than still air

Direct Sunlight — Sunlight adds radiant heat energy directly to the ice cream surface

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

Why does ice cream melt faster on a hot day than on a cool day? Use evidence from your simulation to support your answers.

1 Why does heat always flow from hot to cold and never the other way?

2 How is a thermos like a winter coat? What do they have in common?

3 What surprised you most about how insulation affected melting speed?

4 If you could add one more component to this model, what would it be and why?

5 How does understanding heat transfer help engineers design better products?

8 STEM Challenge: Design the Ultimate Ice Cream Keeper

ModelIt!

YOUR ENGINEERING MISSION

Design an insulated container that keeps ice cream frozen the longest on a hot day.

THE SCENARIO

Your school's summer fair is outdoors in 95-degree heat. The ice cream stand needs a way to keep scoops frozen for at least 30 minutes without electricity. Your team must design and test an insulated container.

Engineering Design Thinking

1. WHAT INSULATING MATERIALS WILL YOU USE (FOAM, FABRIC, FOIL, AIR GAPS)?
2. HOW WILL YOU MINIMIZE THE OPENING WHERE HEAT CAN ENTER?
3. HOW WILL YOU TEST WHETHER YOUR DESIGN ACTUALLY SLOWS MELTING?
4. HOW WILL YOU MEASURE MELTING SPEED TO COMPARE DESIGNS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Materials Scientists and Thermal Engineers design insulation for everything from food containers to spacecraft. They use the same heat transfer principles you modeled today. They earn $75,000-$130,000/year.